Quiz-2

1. Partial Derivatives/Chain Rule.
2. Maximum/Minimum (Extremum for multivariable functions)
3. Taylor Series

Date: 30-04-2026.

Critical point: $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$ /

1st partial derivative of $f(x_0, y_0)$ doesnot exist.

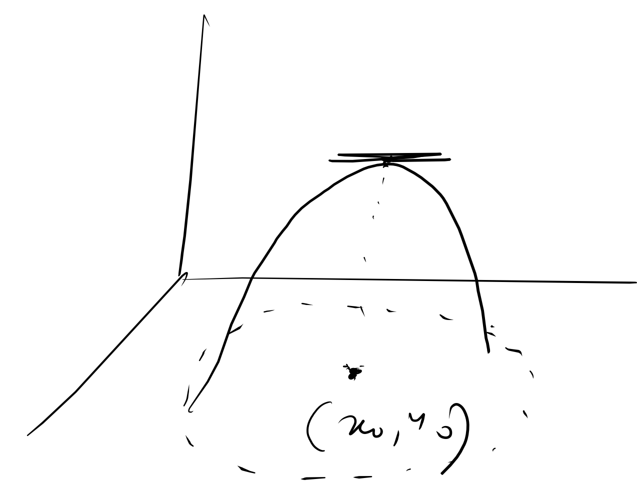
2nd Derivative Test:

(i) $f''(x_0) > 0 \rightarrow$ x_0 minimum domain

$f(x_0)$ minimum function point

(ii) $f''(x_0) < 0 \rightarrow$ $x_0 \rightarrow$ maximum Domain

$f(x_0)$ maximum function point



let $f(x,y)$ be function of two variables with continuous second order partial Derivatives in some open disk centered at critical point (x_0, y_0).

let $D = \begin{vmatrix} f_{xx}(x_0,y_0) & f_{xy}(x_0,y_0) \\ f_{xy}(x_0,y_0) & f_{yy}(x_0,y_0) \end{vmatrix}$ ← x $\left[f_{xy}(x_0,y_0) = f_{yx}(x_0,y_0) \right]$

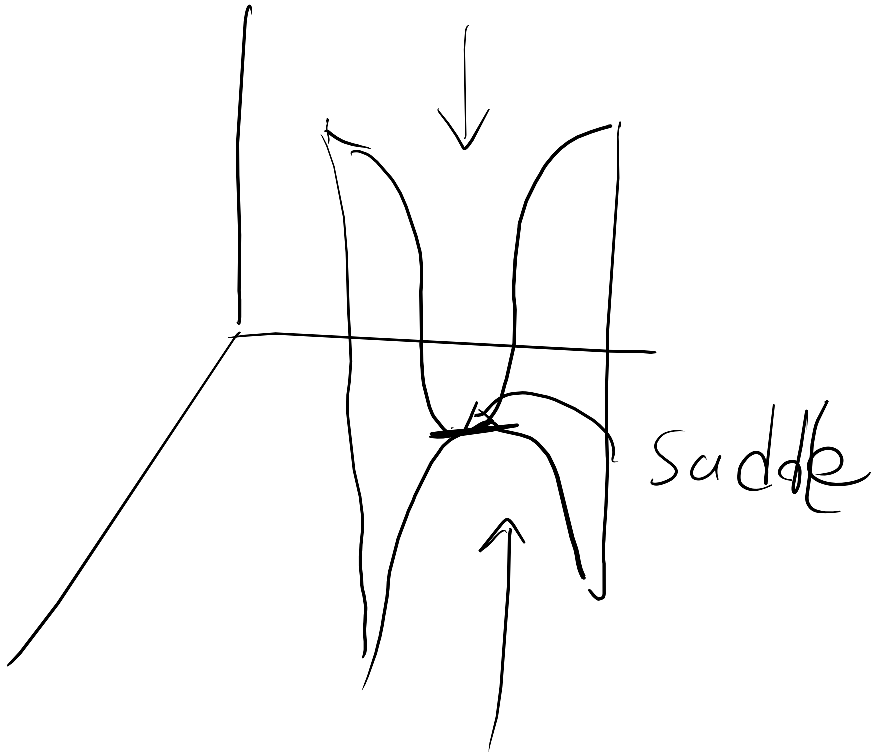
$$= f_{xx}(x_0,y_0)\, f_{yy}(x_0,y_0) - \left(f_{xy}(x_0,y_0)\right)^2.$$

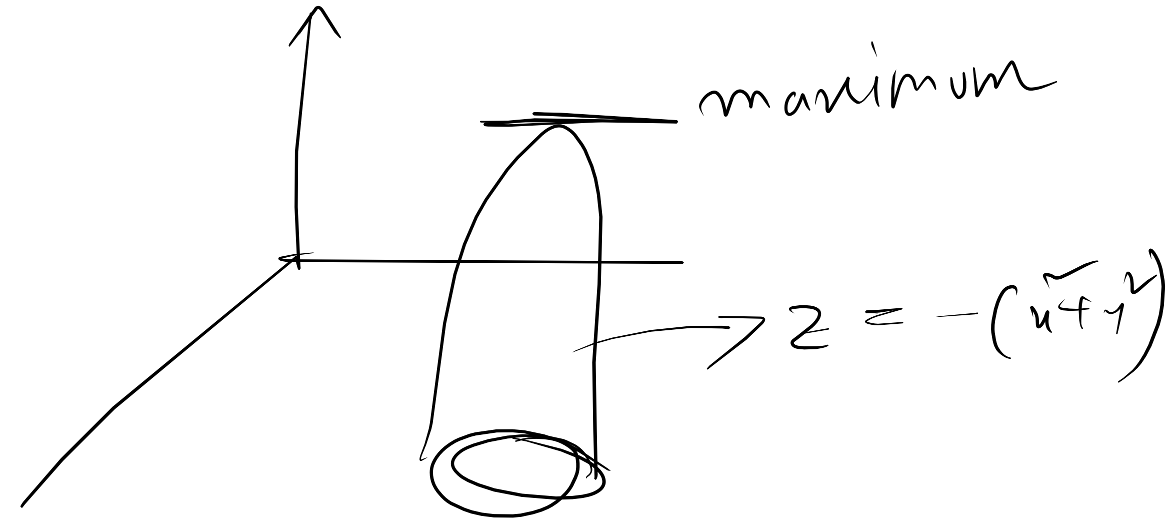
(i) If $D > 0$ & $f_{xx}(x_0,y_0) > 0$, f has relative minimum at (x_0,y_0).

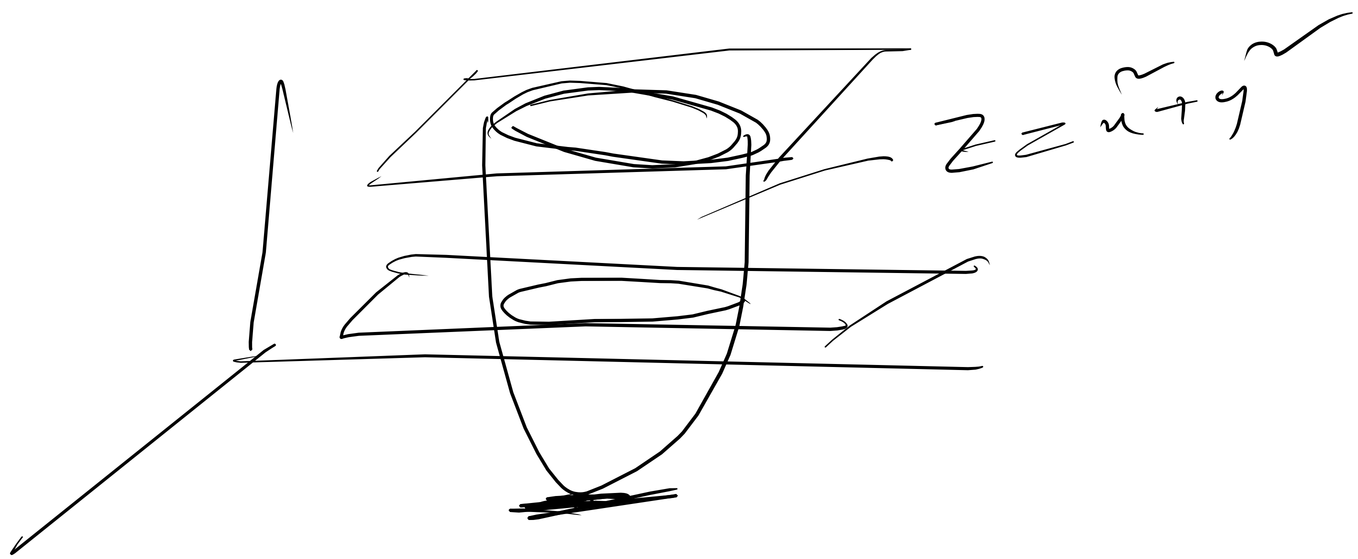
(ii) If $D > 0$, & $f_{xx}(x_0,y_0) < 0$, $f(x_0,y_0)$ has relative maximum at (x_0,y_0)

(iii) If $D < 0$, f has a saddle point at (x_0,y_0)

(iv) If $D = 0$, no conclusion can be drawn


saddle


maximum
z = -(x^2+y^2)


z = x^2+y^2

Ex: 01 Locate all relative maxima, relative minima & saddle points

for $f(x,y) = xy + \frac{2}{x} + \frac{4}{y} \longrightarrow$ Given

$f_x(x,y) = y - \frac{2}{x^2} + 0$ $\qquad f_{xx} = \frac{4}{x^3}$

$f_y(x,y) = x - \frac{4}{y^2}$ $\qquad f_{yy} = \frac{8}{y^3}$ $\qquad f_{xy} = 1.$

for critical points;

$f_x(x,y) = 0$ $\qquad f_y(x,y) = 0$

$\Rightarrow y - \frac{2}{x^2} = 0$ $\qquad \Rightarrow x = \frac{4}{y^2}$ ---- (ii)

$\Rightarrow y = \frac{2}{x^2}$ ---- (i)

Now from (ii) $\Rightarrow x = \frac{4}{y^2}$

$\Rightarrow x = \frac{4}{\left(\frac{2}{x^2}\right)^2}$

$\Rightarrow x = \frac{4}{\frac{4}{x^4}}$

$\Rightarrow x = x^4$

$\Rightarrow x^4 - x = 0$

$\Rightarrow x(x^3 - 1) = 0$

$x = 0 \quad x^3 - 1 = 0$

$x^3 - 1^3 = 0 \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

$\Rightarrow (x-1)(x^2 + x + 1) = 0$

$x - 1 = 0$

$\Rightarrow x = 1,$

$x \in \mathbb{C}$

if $x = 0$, from (i)

$y = \frac{2}{0^2} = \underline{\text{undefined}}$

$x = 1, \; y = 2$

$(1, 2)$ is critical point.

$$D = f_{xx}(1,2)\, f_{yy}(1,2) - (f_{xy}(1,2))^2$$

$$= \left(\frac{4}{1^3}\right)\left(\cancel{\frac{8}{2^3}}\right) - 1^2$$

$$f_{xx} = \frac{4}{x^3}$$

$$f_{xx}(1,2) = \frac{4}{1^3} = 4.$$

$$= 3 > 0, \quad f_{xx}(1,2) = 4 > 0$$

Relative minimum point.

$f(1,2) = 1 \cdot 2 + \frac{2}{1} + \frac{4}{2} = 6$ is a minimum point of f.

Ex-02 $f(x,y) = 4xy - x^4 - y^4$.

$f_x = 4y - 4x^3$. $f_{xx} = -12x^2$, $f_{yy} = -12y^2$,

$f_y = 4x - 4y^3$ $f_{xy} = 4$.

Critical points:

$f_x = 0$

$\Rightarrow 4y - 4x^3 = 0$

$\Rightarrow \boxed{y = x^3}$ . . . (1)

$f_y = 0$

$\Rightarrow 4x - 4y^3 = 0$

$\Rightarrow x = y^3$ . . . (11)

from (11) $x = (x^3)^3$

$\Rightarrow x^9 = x$

$\Rightarrow x^9 - x = 0$

$\Rightarrow x(x^8 - 1) = 0$

$x = 0$, $x^8 - 1 = 0$

$\underline{x=0}$, $x^8-1=0$

$\Rightarrow (x^4)^2-1^2=0$

$\Rightarrow \underbrace{(x^4+1)}(x^4-1)=0$

$\Rightarrow x^4-1=0$

$\Rightarrow (x^2)^2-1=0$

$\Rightarrow \underbrace{(x^2+1)}(x^2-1)=0$

$x=\pm 1.$

from (1), $y=x^3$

$x=0,\ y=0$

$x=1,\ y=(1)^3=1$

$x=-1,\ y=(-1)^3=-1.$

$\boxed{x^4=-1}$

$\boxed{x^2+1=0}$

$x^4+1=0$

$\Rightarrow \underline{x\in \mathbb{C}}$

$(0,0)$, $(1,1)$, $(-1,-1)$ are critical points.

$$D = f_{xx} f_{yy} - f_{xy}^2$$
$$= (-12x^2)(-12y^2) - 4^2 = 144x^2y^2 - 16.$$

(i) $D(0,0) = -16 < 0$, $f(0,0) = 0$ is a saddle point of f.

(ii) $D(1,1) = 144 - 16 = 128 > 0$, $f_{xx}(1,1) = -12 < 0$,

$f(1,1) = 4 - 1 - 1 = 2$ is a relative maximum. of f

(iii) $D(-1,-1) = 144 - 16 = 128 > 0$, $f_{xx}(-1,-1) = -12 < 0$

$f(-1,-1) = 2$ is relative maximum of f.

1. $f(x,y) = x^2 + xy - 2y - 2x + 1$.

2. $f(x,y) = xy - x^3 - y^2$.

3. $f(x,y) = x^3 + y^3 - 3xy + y$

Anton 13.8

(i) Example

(ii) Exercise (9-19).